

# Activity 1: Reading a Price Chart

Build two price-time charts from real market data, identify what the record says, and then forecast what it does not.

## Student worksheet

20 points | about 90 minutes

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Name

Section

Date

## Before you start

What this is, what it is worth, and what is actually being marked.

Chapter 1 says technical analysis does exactly two jobs. **Identification** is the descriptive job: it records and describes past and present price and market action. **Forecasting** is the inferential job: it interprets what was identified to infer what price might do next.

The difference between them is the whole of this activity. An identification can be marked right or wrong, because the record either says that or it does not. A forecast cannot be marked right or wrong at the moment you write it, because nobody has seen what happens next. You will do both, and you will be marked on them differently.

### READ THIS TWICE

**Your forecast in Part B is not marked right or wrong.** It is marked on whether the reasons you give are things your own chart actually shows. Looking up what the price did next will not earn you a single mark, and copying it will lose you the reasoning marks. In the session after this is due you will be shown what price really did, and the point of that reveal is not to find out who was lucky.

## What you need

- Any computer, tablet or phone with a web browser. A borrowed one is fine.
- A free Google account. Nothing to install and nothing to pay for.
- About 90 minutes. Part A takes the longest because every click is spelled out.
- A printer, or somebody with one, for the two charts you draw on by hand.

The activity is built in Google Sheets because the screen looks the same for everyone, which is what lets this sheet show you a picture of every step. **Everything you do here transfers to Microsoft Excel.** The menus sit in different places and the formulas are identical.

## Where the numbers come from

Both price series are published by **FRED**, the economic database of the Federal Reserve Bank of St. Louis, and both are measured by the **U.S. Energy Information Administration**. FRED marks them public domain, which is why this course can hand them to you rather than ask you to hunt for data of your own.

|                      | PART A                                       | PART B                                                          |
|----------------------|----------------------------------------------|-----------------------------------------------------------------|
| What it is           | Henry Hub natural gas spot price             | West Texas Intermediate crude oil spot price, Cushing, Oklahoma |
| FRED code            | DHHNGSP                                      | DCOILWTICO                                                      |
| Price is measured in | US dollars per million British thermal units | US dollars per barrel                                           |
| First trading day    | 1 November 2013                              | 3 February 2014                                                 |
| Last trading day     | 31 March 2014                                | 30 June 2014                                                    |
| Trading days         | 102                                          | 103                                                             |

Retrieved from FRED on 4 September 2026. The only change made to either file was to remove the days the market did not trade, which FRED leaves in the file with the price blank: 5 of them in Part A and 3 in Part B, all of them United States public holidays. No price was altered, added or removed.

#### CLOSING PRICES ONLY

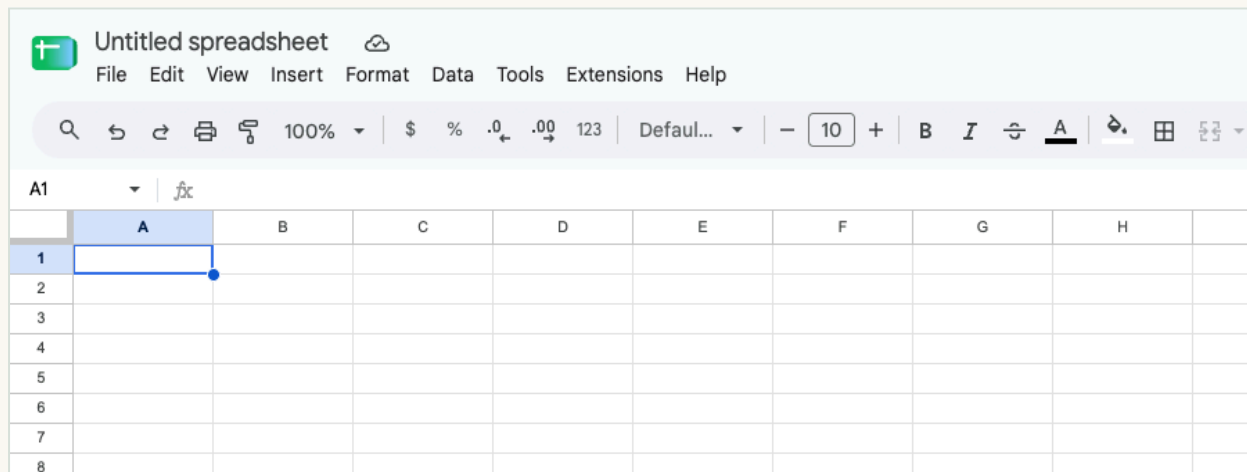
These series carry one price per day, the closing price. They carry no opening price, no high, no low and no volume, so this activity never asks you for any of those. Chapter 1 calls the full set of four **OHLC**, and you will meet it properly when we get to bar charts.

## Part A. Build it once, with every click shown

Henry Hub natural gas spot price, 1 November 2013 to 31 March 2014. Follow the steps in order. Do not skip one because it looks obvious.

### 1 Open a new, empty spreadsheet

In your browser's address bar type `sheets.new` and press Enter. Sign in to your Google account if you are asked to. An empty spreadsheet opens, called Untitled spreadsheet.

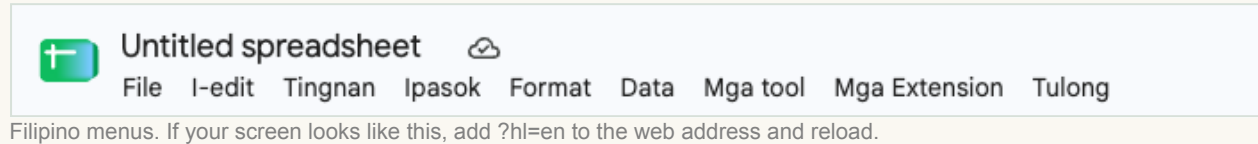


An empty Google Sheet. The grid of lettered columns and numbered rows is where the prices will land.

## 2 Make sure your menus are in English

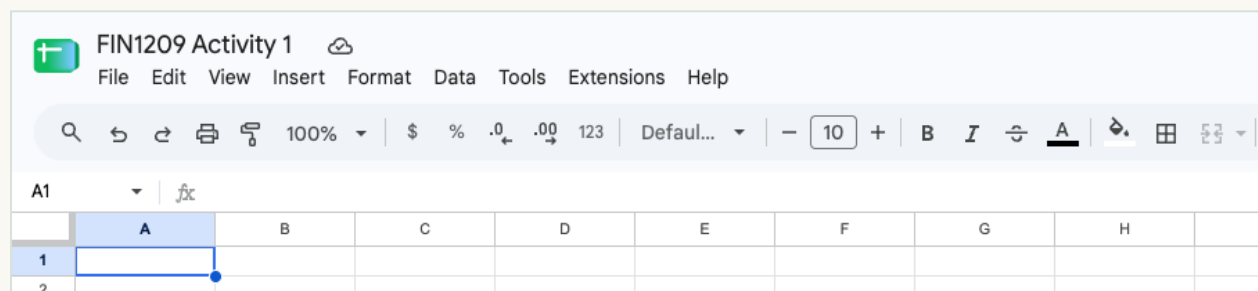
Look along the top. It must read File, Edit, View, Insert, Format, Data, Tools, Extensions, Help. If it reads File, I-edit, Tingnan, Ipasok, Format, Data, Mga tool, Mga Extension, Tulong, Google has decided you want Filipino and the rest of this sheet will not match your screen.

To fix it, click at the end of the web address at the very top of your browser, add `?hl=en` to it, and press Enter. The page reloads in English. Do this now, before anything else.



## 3 Give the file its name

Click the words 'Untitled spreadsheet' at the top left, type `FIN1209 Activity 1` and press Enter. Do not put your own name in the file name. You write your name on this printed sheet.



The file is named. It saves itself, so there is no save button to look for.

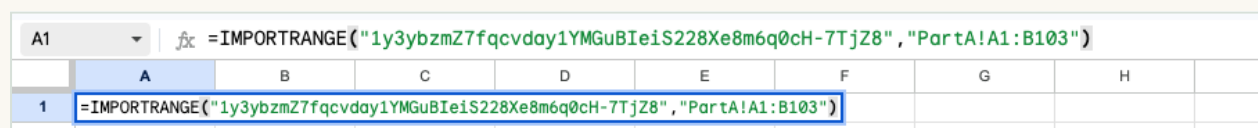
## 4 Click cell A1, then type in the formula that fetches the prices

Click the very first cell, the one where column A meets row 1. Then type this, exactly, including every bracket and quotation mark. It is one line. Do not press Enter yet.

TYPE EXACTLY THIS

```
=IMPORTRANGE("1y3ybmZ7fqcvday1YMGuBIeIS228Xe8m6q0cH-7TjZ8","PartA!A1:B103")
```

`IMPORTRANGE` reaches into the spreadsheet your instructor shared and copies a block of it into yours. The long string of letters is that spreadsheet's address. `PartA!A1:B103` is the part being copied: the tab called PartA, from cell A1 down to cell B103.



The formula sits in cell A1 and is repeated in the formula bar above the grid. Check it character by character before you go on.

## 5 Press Enter and watch the prices arrive

Column A fills with dates and column B with prices, 102 rows of them. Row 1 holds the headers that came with the file: `observation_date` and `DHHNGSP`, which is FRED's own code for this price. Leave them alone. They are the proof of where your numbers came from.

If instead you see `#REF!`, you have mistyped the long address. Click A1, compare it character by character with the box above, and fix it. If a message appears asking you to connect the two sheets, click Allow access.

|    | A                | B       | C | D | E | F | G |
|----|------------------|---------|---|---|---|---|---|
| 1  | observation_date | DHHNGSP |   |   |   |   |   |
| 2  | 2013-11-01       | 3.45    |   |   |   |   |   |
| 3  | 2013-11-04       | 3.38    |   |   |   |   |   |
| 4  | 2013-11-05       | 3.37    |   |   |   |   |   |
| 5  | 2013-11-06       | 3.46    |   |   |   |   |   |
| 6  | 2013-11-07       | 3.59    |   |   |   |   |   |
| 7  | 2013-11-08       | 3.54    |   |   |   |   |   |
| 8  | 2013-11-11       | 3.62    |   |   |   |   |   |
| 9  | 2013-11-12       | 3.69    |   |   |   |   |   |
| 10 | 2013-11-13       | 3.69    |   |   |   |   |   |
| 11 | 2013-11-14       | 3.52    |   |   |   |   |   |
| 12 | 2013-11-15       | 3.56    |   |   |   |   |   |
| 13 | 2013-11-18       | 3.71    |   |   |   |   |   |
| 14 | 2013-11-19       | 3.63    |   |   |   |   |   |
| 15 | 2013-11-20       | 3.62    |   |   |   |   |   |

Dates down column A, closing prices down column B.

## 6 Check that all of it arrived

Click the small white box at the top left that currently says A1. That is the Name Box. Type `B103` into it and press Enter. The spreadsheet jumps to the last price.

The Name Box now reads `B103` and the formula bar beside it reads `4.48`. Row 103 shows `2014-03-31`. If yours says anything else, your range is wrong. Go back to step 4.

|    | A          | B    | C | D | E | F | G |
|----|------------|------|---|---|---|---|---|
| 72 | 2014-02-13 | 5.34 |   |   |   |   |   |

The Name Box and the formula bar are how you check any cell without hunting for it.

## 7 Select every date and every price

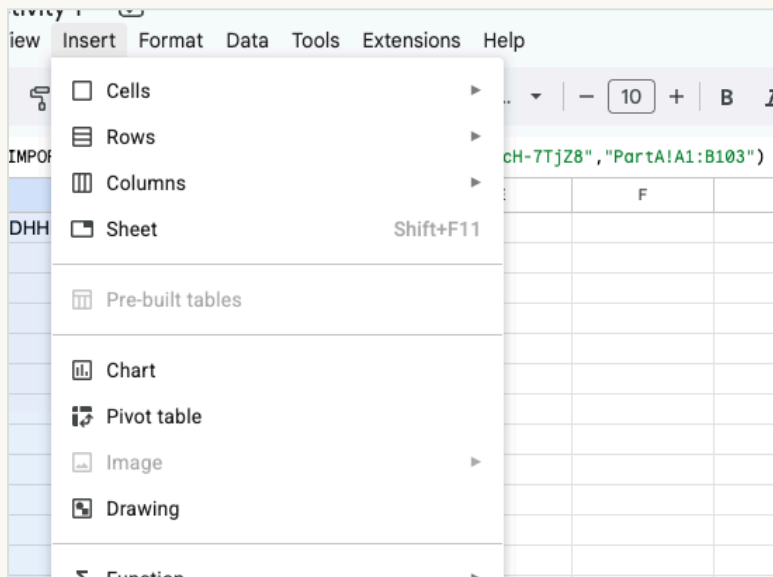
Type `A1:B103` into the Name Box and press Enter. Columns A and B go blue from row 1 to row 103. Down in the bottom right corner the sheet counts what you have selected: 206 cells, which is 102 dates plus 102 prices plus the two headers.

|   | A                | B       | C | D | E | F | G | H |  |
|---|------------------|---------|---|---|---|---|---|---|--|
| 1 | observation_date | DHHNGSP |   |   |   |   |   |   |  |
| 2 | 2013-11-01       | 3.45    |   |   |   |   |   |   |  |
| 3 | 2013-11-04       | 3.38    |   |   |   |   |   |   |  |
| 4 | 2013-11-05       | 3.37    |   |   |   |   |   |   |  |
| 5 | 2013-11-06       | 3.46    |   |   |   |   |   |   |  |
| 6 | 2013-11-07       | 3.59    |   |   |   |   |   |   |  |
| 7 | 2013-11-08       | 3.54    |   |   |   |   |   |   |  |

The Name Box selects a whole block in one move, which is faster and far more reliable than dragging.

## 8 Insert the chart

With that block still selected, click **Insert** in the menu bar and then **Chart**.



Insert, then Chart.

## 9 Look at what you have made

A chart appears, and a Chart editor panel opens on the right. Google has already worked out that dates along the bottom and prices up the side make a line chart, so it has drawn one.

This is the object Chapter 1 calls a **price-time chart**: price on the vertical axis, time on the horizontal axis, one price line plotted against both. Everything else in this course is drawn on top of this.



Your first price-time chart, built from a column of numbers you did not type.

## 10 Open the titling controls

In the Chart editor on the right, click **Customize**, then click **Chart and axis titles** to open it.

The image shows the 'Customize' tab of the Google Charts editor. The 'Chart style' section is expanded, and the 'Chart & axis titles' subsection is open. It contains a 'Chart title' dropdown menu with a green arrow, a 'Title text' input field containing 'DHHNGSP vs. observation\_date', and 'Title font' and 'Title font size' labels at the bottom.

The Customize tab, with Chart and axis titles open.

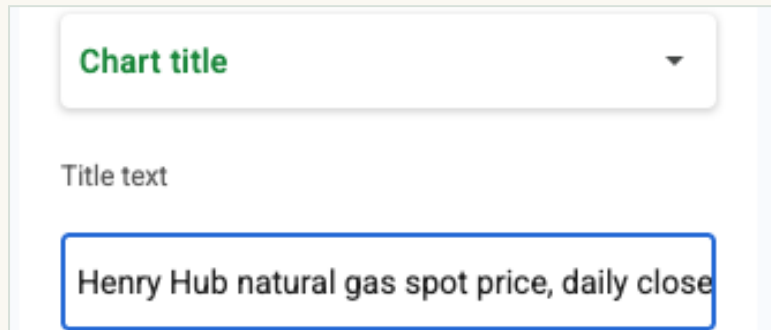
## 11 Title the chart

The box under Title text holds Google's guess. Delete it and type this instead.

TYPE EXACTLY THIS

Henry Hub natural gas spot price, daily close

The title on the chart changes as you type.

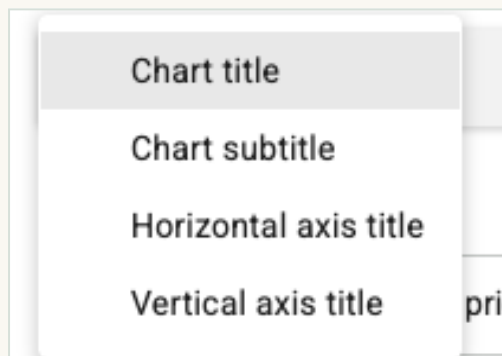


The screenshot shows the Google Charts configuration interface. At the top, there is a dropdown menu labeled 'Chart title' with a downward arrow. Below it, there is a section labeled 'Title text' which contains a text input box. The input box is highlighted with a blue border and contains the text 'Henry Hub natural gas spot price, daily close'.

Title text, typed in. Nothing needs saving.

## 12 Switch from the chart title to the axis titles

Click the dropdown that currently says Chart title. Four choices appear. Choose **Horizontal axis title**, and in the Title text box type `Time`.



The screenshot shows the dropdown menu that appears when clicking the 'Chart title' dropdown. The menu has four options: 'Chart title', 'Chart subtitle', 'Horizontal axis title', and 'Vertical axis title'. The 'Horizontal axis title' option is highlighted with a grey background.

The same dropdown gives you the chart title, the subtitle and both axis titles.

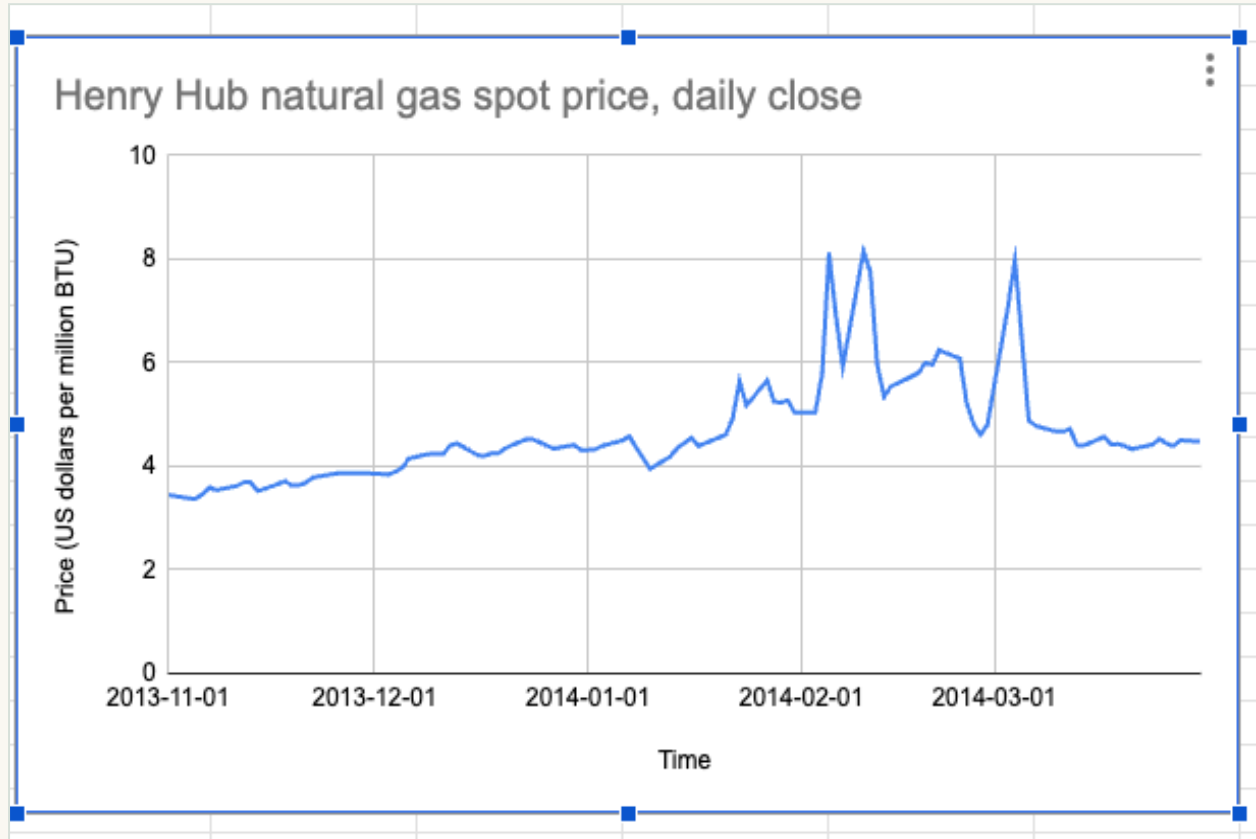
### 13 Now name the other axis

Open that dropdown again, choose **Vertical axis title**, and type this into the Title text box.

TYPE EXACTLY THIS

Price (US dollars per million BTU)

Both axes are now named. You have just labelled the two dimensions of the price-time chart by hand, which is worth doing once so that you never again look at a chart without asking what its axes are.

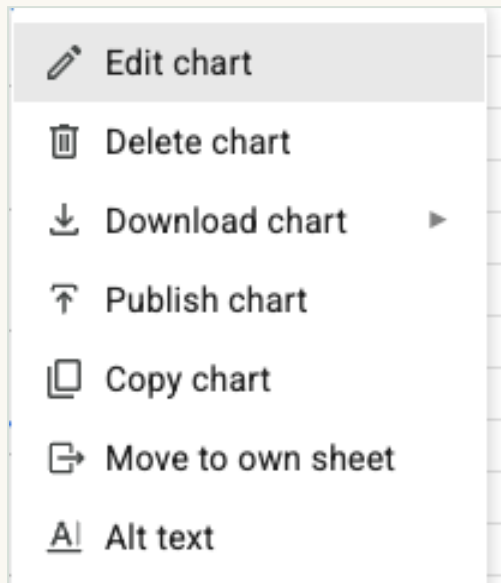


Price up the side, time along the bottom, both labelled.



#### 14 Give the chart a page of its own

Click the three dots at the top right corner of the chart itself, not the editor. Choose **Move to own sheet**.

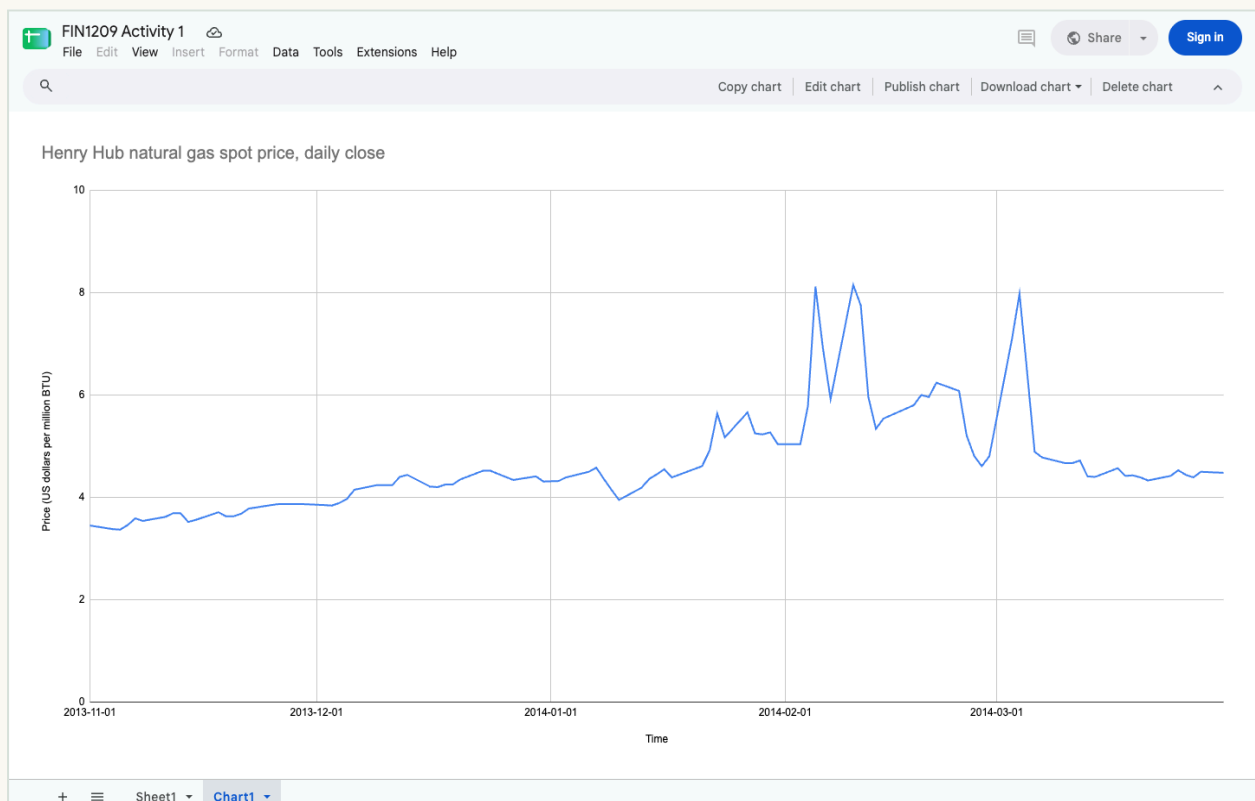


The chart's own menu. Move to own sheet is near the bottom.

#### 15 This is the chart you print

The chart now fills a tab of its own, called Chart1, and the tabs along the bottom let you move between it and your numbers. It is big enough to read and big enough to draw on.

Print this tab, or take a screenshot of it. You need it on paper later, and you need the Part B one on paper too.



The finished Part A chart on its own tab.

## 16 Go back to the numbers and start a third column

Click the **Sheet1** tab at the bottom. Using the Name Box, go to cell **C1** and type **Daily change**. Then go to cell **C3** and type this.

TYPE EXACTLY THIS

=B3-B2

C3 now shows how much the price moved between the first day and the second. C2 stays empty on purpose: the first day has no day before it to be compared with.

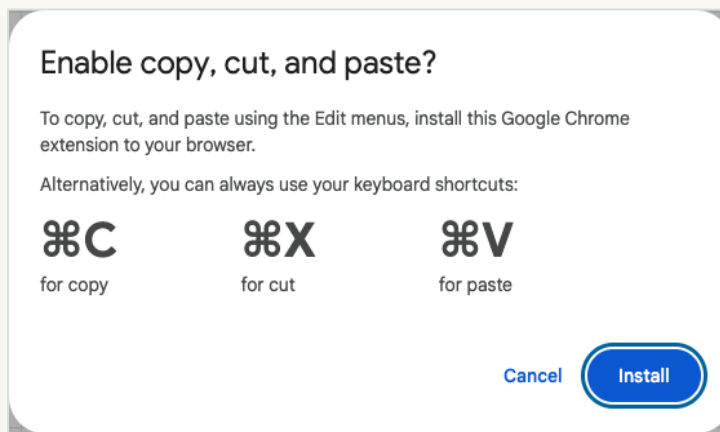
| C3 | A                | B       | C            | D | E | F |
|----|------------------|---------|--------------|---|---|---|
| 1  | observation_date | DHHNGSP | Daily change |   |   |   |
| 2  | 2013-11-01       | 3.45    |              |   |   |   |
| 3  | 2013-11-04       | 3.38    | -0.07        |   |   |   |
| 4  | 2013-11-05       | 3.37    |              |   |   |   |

One day's change, in one cell.

## 17 Copy that formula down all the way

Click C3. Press **Ctrl and C** together to copy it, or **Command and C** on a Mac. Then type **C4:C103** into the Name Box, press Enter, and press **Ctrl and V**, or **Command and V** on a Mac, to paste.

The spreadsheet adjusts the formula for every row as it pastes: C4 becomes **=B4-B3**, C5 becomes **=B5-B4**, and so on down to C103. This is the single most useful thing a spreadsheet does. Do not use the Edit menu for this: it asks you to install an extension. Use the keyboard.



What the Edit menu offers instead. If you see this, click Cancel and use the keyboard.

## 18 Check the column filled correctly

Go to **C103** in the Name Box. The formula bar should read **=B103-B102**. Column C is now a record of every single day's move, in the same units as the price.

|    | A                | B      | C            | D |
|----|------------------|--------|--------------|---|
| 1  | observation_date | DHNGSP | Daily change |   |
| 2  | 2013-11-01       | 3.45   |              |   |
| 3  | 2013-11-04       | 3.38   | -0.07        |   |
| 4  | 2013-11-05       | 3.37   | -0.01        |   |
| 5  | 2013-11-06       | 3.46   | 0.09         |   |
| 6  | 2013-11-07       | 3.59   | 0.13         |   |
| 7  | 2013-11-08       | 3.54   | -0.05        |   |
| 8  | 2013-11-11       | 3.62   | 0.08         |   |
| 9  | 2013-11-12       | 3.69   | 0.07         |   |
| 10 | 2013-11-13       | 3.69   | 0            |   |
| 11 | 2013-11-14       | 3.52   | -0.17        |   |
| 12 | 2013-11-15       | 3.56   | 0.04         |   |
| 13 | 2013-11-18       | 3.71   | 0.15         |   |
| 14 | 2013-11-19       | 3.63   | -0.08        |   |
| 15 | 2013-11-20       | 3.63   | 0            |   |
| 16 | 2013-11-21       | 3.68   | 0.05         |   |
| 17 | 2013-11-22       | 3.78   | 0.1          |   |
| 18 | 2013-11-25       | 3.85   | 0.07         |   |
| 19 | 2013-11-26       | 3.87   | 0.02         |   |

Dates, prices and daily changes, side by side.

## 19 Set up the answer block

Go to cell **E1** and type these sixteen labels, pressing Enter after each one so you move down the column: Trading days, First close, Last close, Change over the window, Percent change, Highest close, Date of highest close, Lowest close, Date of lowest close, Days price rose, Days price fell, Days price did not move, Biggest one day rise, Date of biggest rise, Biggest one day fall, Date of biggest fall.

They will look as though they spill into column F. They do not: column F is empty, so the text simply shows across it. Leave column F alone.

## 20 Type the sixteen formulas

Now go to cell **G1** and type the formula from the first row of the table below. Press Enter, which moves you to G2, and type the second. Carry on to G16. Every one of them is a question about the record, and every answer is a fact you could not argue with.

**MAX** and **MIN** find the largest and smallest number in a block. **COUNT** counts how many numbers are in it. **COUNTIF** counts only the ones that pass a test. **INDEX** with **MATCH** is the pair that answers **when**: **MATCH** finds which row a value sits on, and **INDEX** reads the date off that same row.

|  | E                      | F | G    |  |
|--|------------------------|---|------|--|
|  | Trading days           |   | 102  |  |
|  | First close            |   | 3.45 |  |
|  | Last close             |   | 4.48 |  |
|  | Change over the window |   | 1.03 |  |

The top of your answer block. Check these three against your own screen before you type the rest. The other thirteen answers are yours to produce.

### The sixteen formulas, in order, down column G

| CELL | TYPE THIS                               |
|------|-----------------------------------------|
| G1   | =COUNT (B2:B103)                        |
| G2   | =B2                                     |
| G3   | =B103                                   |
| G4   | =G3-G2                                  |
| G5   | =ROUND (G4/G2*100,2)                    |
| G6   | =MAX (B2:B103)                          |
| G7   | =INDEX (A2:A103,MATCH (G6,B2:B103,0) )  |
| G8   | =MIN (B2:B103)                          |
| G9   | =INDEX (A2:A103,MATCH (G8,B2:B103,0) )  |
| G10  | =COUNTIF (C3:C103,">0")                 |
| G11  | =COUNTIF (C3:C103,"<0")                 |
| G12  | =COUNTIF (C3:C103,"=0")                 |
| G13  | =MAX (C3:C103)                          |
| G14  | =INDEX (A3:A103,MATCH (G13,C3:C103,0) ) |
| G15  | =MIN (C3:C103)                          |
| G16  | =INDEX (A3:A103,MATCH (G15,C3:C103,0) ) |

Each formula sits beside the label you typed into column E on the row above, so G1 answers the label in E1.

### What the record says

Every question below is an act of **identification**. Not one of them is a prediction, an opinion or an interpretation. Each has exactly one right answer, your spreadsheet has already worked it out, and your answer will be marked against it.

## Part A answers: copy them off your own screen

Read each one out of column G of your spreadsheet and write it here. Give prices to two decimal places and dates as they appear.

- |                               |     |       |
|-------------------------------|-----|-------|
| 1. Trading days in the window | G1  | _____ |
| 2. First close                | G2  | _____ |
| 3. Last close                 | G3  | _____ |
| 4. Change over the window     | G4  | _____ |
| 5. Percent change             | G5  | _____ |
| 6. Highest close              | G6  | _____ |
| 7. Date of the highest close  | G7  | _____ |
| 8. Lowest close               | G8  | _____ |
| 9. Date of the lowest close   | G9  | _____ |
| 10. Days price rose           | G10 | _____ |
| 11. Days price fell           | G11 | _____ |
| 12. Days price did not move   | G12 | _____ |
| 13. Biggest one day rise      | G13 | _____ |
| 14. Date of the biggest rise  | G14 | _____ |
| 15. Biggest one day fall      | G15 | _____ |
| 16. Date of the biggest fall  | G16 | _____ |

## One thing to notice before you go on

Look at your chart again. It is not a trend. Price drifted gently for three months, went almost vertical in February, and came most of the way back down. From the first day to the last it rose 30 percent. At the high on 10 February 2014 it had risen 136 percent, and it gave almost all of that back in seven weeks.

That is worth seeing once. The shape of a chart and the change from the first day to the last are different facts, and a chart that ends near where it began has not necessarily been quiet. Both are identification. Neither is a forecast.

## Part B. Now do it yourself

West Texas Intermediate crude oil spot price, Cushing, Oklahoma, 3 February 2014 to 30 June 2014. The same build, the same sixteen questions, no pictures.

This is a different market: crude oil rather than natural gas, priced in dollars a barrel rather than dollars a million BTU. The method does not change at all, which is the point of doing it twice.

This is the formula for step 2 below. It is the Part A formula with one word and two row numbers changed.

TYPE EXACTLY THIS

```
=IMPORTRANGE ("1y3ybmZ7fqcvday1YMGuBIeiS228Xe8m6q0cH-7TjZ8", "PartB!A1:B104")
```

## Build it

1. In the same spreadsheet, click the plus sign at the bottom left to add a new tab. It appears as Sheet2.
2. Click cell A1 of the new tab and type the formula in the box above.
3. Check cell B104. It should read 106.07, on 2014-06-30.
4. Select A1:B104 with the Name Box, then Insert and Chart. Title it, name both axes, and move it to its own sheet, exactly as in steps 10 to 15. The vertical axis is Price (US dollars per barrel) this time.
5. Build column C the same way: Daily change in C1, =B3-B2 in C3, copied down to C104.
6. Build the same sixteen labels in column E and the same sixteen formulas in column G, changing every 103 to 104 and every 102 to 103.
7. Print your Part B chart, or take a screenshot of it and print that. You are about to draw on it.

## Part B answers

The same sixteen questions, about a different market. Read them off column G of your Part B tab.

- |                               |     |       |
|-------------------------------|-----|-------|
| 1. Trading days in the window | G1  | _____ |
| 2. First close                | G2  | _____ |
| 3. Last close                 | G3  | _____ |
| 4. Change over the window     | G4  | _____ |
| 5. Percent change             | G5  | _____ |
| 6. Highest close              | G6  | _____ |
| 7. Date of the highest close  | G7  | _____ |
| 8. Lowest close               | G8  | _____ |
| 9. Date of the lowest close   | G9  | _____ |
| 10. Days price rose           | G10 | _____ |
| 11. Days price fell           | G11 | _____ |
| 12. Days price did not move   | G12 | _____ |
| 13. Biggest one day rise      | G13 | _____ |
| 14. Date of the biggest rise  | G14 | _____ |
| 15. Biggest one day fall      | G15 | _____ |
| 16. Date of the biggest fall  | G16 | _____ |

## Mark the peaks and the troughs

On your printed Part B chart, in pen, mark every point where the price line turned. A **peak** is a point with lower prices on both sides of it. A **trough** is a point with higher prices on both sides. Write a P beside each peak and a T beside each trough.

Chapter 1 gives the most widely accepted definition of a trend as **successively higher or lower peaks and troughs**. You now have the marks to test that definition against, so answer the two questions below from your own marked chart.

## Are your troughs getting higher, lower, or neither?

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## Are your peaks getting higher, lower, or neither?

## EXPECT TO DISAGREE

The person beside you will not mark the same peaks and troughs you did, and neither of you will have made a mistake. Chapter 1 calls that **subjectivity**: two competent analysts, given the same chart, reaching different and equally defensible conclusions. You met it in class with the book's USDCAD chart. Here it is again, with your own hand doing the drawing.

## Draw one trendline

Take a ruler. Draw **one** straight line that touches at least two of your troughs and that no part of the price line drops below between them. Extend that line past the right hand edge of the chart, into the empty space where the future would be.

Label the two troughs your line rests on, and write beside the line the price it would sit at one month after the chart ends. You do not need to be exact. Read it off your own extended line.

**The two dates your line touches, and where it sits one month after the chart ends**

## Now, and only now, forecast

Everything up to this point has been identification. This part is not. Write five to eight sentences saying what you think the price does next, and why.

**You are marked on the reasons, not on the answer.** A forecast that turns out wrong and is argued from three things visible on your chart earns full marks. A forecast that turns out right with no reasons earns almost none. Say which way you think price goes, name the marks on your own chart that make you think so, and name the one thing you can see that argues against you.

## Last step. Five minutes on a real charting platform

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Because the department's version of this homework asks for one, and because you should see one before the next chapter.

A spreadsheet is where you learn what a chart is made of. A charting platform is where the work is actually done. Open one, find a chart, draw a line on it, and stop.

1. Go to [www.tradingview.com/chart/](https://www.tradingview.com/chart/). It is free and you do not have to sign in to draw on a chart.
2. Click the symbol name at the top left and search for any market you like. A share, a currency pair, gold, oil, anything.
3. Set the timeframe to daily using the D button at the top, and pick a stretch of chart where price clearly went one way for a while.
4. On the toolbar down the left hand side, the second button from the top is the **Trendline** tool. Click it, then click once at one end of your line and once at the other.
5. Take a screenshot of the result with your line on it.

### ONE SENTENCE, UNDERNEATH

Write one sentence saying what your line touches and what would have to happen for price to break it. That sentence is worth as much as the screenshot.

### Your one sentence

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## What to hand in, and how it is marked

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### Four things, in one submission

1. This sheet, filled in, with your name on the front.
2. Your Part A chart and your Part B chart, printed, with the peaks, the troughs and your one trendline drawn on the Part B one.
3. The link to your spreadsheet, or a screenshot of your Part B tab showing columns A to G.
4. Your TradingView screenshot with its one sentence.



## The 20 marks

| WHAT                             | MARKS | WHAT EARNS THEM                                                                                                                                      |
|----------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part A, built                    | 2     | The chart exists, has a title and two named axes, and the prices are the right ones.                                                                 |
| Part A, identified               | 6     | Sixteen answers, half a mark each, rounded down. Right or wrong against the record.                                                                  |
| Part B, built                    | 2     | The same, done without the pictures.                                                                                                                 |
| Part B, identified               | 6     | The same sixteen questions, half a mark each, rounded down.                                                                                          |
| Peaks, troughs and one trendline | 2     | Marked on the printed chart, in pen. The line touches at least two troughs and is extended past the edge.                                            |
| The forecast                     | 1     | Five to eight sentences that name at least two things visible on your own chart, and one thing that argues against you. Never marked right or wrong. |
| The platform step                | 1     | A screenshot of a chart you found, with a trendline you drew, and one sentence about it.                                                             |

20 in total. 16 of them are for work that can be marked right or wrong against the record. 4 are not, and those are marked on the reasoning.

### WHAT HAPPENS NEXT

The Part B window stops on a day chosen before you were given it, and the price kept going after that day. In the session after this is due you will be shown exactly what it did. Your identification answers will be marked right or wrong. Your forecast will only be compared. That difference is the whole of Chapter 1's dual function, and it is why this activity exists.